

Statistical Methods in Chemistry

Supplement Block

Significant Figures and Rounding

Student worksheet

Name: _____

Class: _____

Date: _____

Contents

1	Why significant figures matter in chemistry	2
2	Counting significant figures	2
3	Significant figures and decimal places are different	3
4	Addition and subtraction	4
5	Multiplication and division	5
6	Do not round too early	5
7	Exact numbers do not limit significant figures	6
8	Using uncertainty to choose significant figures	7
9	pH and logarithms	8
10	Choosing significant figures in written conclusions	9
11	Mini-practical: reporting a titration result	9
	Answer key	11

1 Why significant figures matter in chemistry

A measured value is not just a number. It is a statement about the quality of a measurement. In chemistry, the number of digits you write should match what the instrument and method can reasonably support.

Core idea

Significant figures are the meaningful digits in a measured or calculated value. They communicate precision. Too few digits throw away useful information; too many digits invent precision that the experiment did not have.

For example, a titre of

$$24.65 \text{ cm}^3$$

communicates a more precise measurement than

$$24.6 \text{ cm}^3.$$

But a value such as

$$24.6537 \text{ cm}^3$$

is normally fake precision for a school burette. The apparatus does not justify that many decimal places.

Do not confuse calculator digits with experimental precision

A calculator may display ten digits. That does not mean the experiment produced ten meaningful digits. The final answer must be rounded according to the measurements and their uncertainties.

Quick exercise: Precision claim

For each value, say whether the reported precision is plausible for the instrument.

Instrument	Reported value	Plausible? Explain briefly.
School burette	24.65 cm^3	
Measuring cylinder marked every 1 cm^3	23.47 cm^3	
Balance reading to 0.001 g	2.347 g	
pH meter displaying two decimals	$\text{pH} = 4.126$	

2 Counting significant figures

Basic rules

- 1) All non-zero digits are significant.
- 2) Zeros between non-zero digits are significant.
- 3) Leading zeros are not significant; they only locate the decimal point.
- 4) Trailing zeros after a decimal point are significant.
- 5) Trailing zeros in whole numbers are ambiguous unless scientific notation is used.

Value	Significant figures	Reason
4.27	3	all digits are non-zero
2.006	4	the zeros are between non-zero digits
0.00450	3	leading zeros do not count; final zero after decimal counts
12.030	5	captive zero and final decimal zero both count
1000	ambiguous	could mean 1, 2, 3, or 4 significant figures
1.000×10^3	4	scientific notation removes ambiguity

Worked example: Trailing zeros

The value 0.1050 has four significant figures: 1, 0, 5, and the final 0. The first zero before the decimal point does not count.

0.1050 has 4 significant figures.

The value 0.105 has only three significant figures. These two values do not communicate the same precision.

Quick exercise: Count the significant figures

Complete the table.

Value	Significant figures	Value	Significant figures
0.00320	_____	40.05	_____
2500	_____	2.500×10^3	_____
0.1000	_____	100.0	_____
7.080	_____	6.02×10^{23}	_____

3 Significant figures and decimal places are different

Significant figures count meaningful digits. Decimal places count digits after the decimal point. A value can have many decimal places but only a few significant figures.

Value	Significant figures	Decimal places
24.65	4	2
0.105	3	3
0.00450	3	5
100.0	4	1
2.0	2	1

This distinction is important because addition/subtraction and multiplication/division use different rounding logic.

Reporting rule: Two different rounding rules

- For **addition and subtraction**, round to the least precise decimal place.
- For **multiplication and division**, round to the lowest number of significant figures among the measured values.

Quick exercise: Significant figures or decimal places?

For each calculation, state which rule should control the final rounding.

- a) $12.45 + 3.2 + 0.567$

b) $0.1000 \times 24.65 / 25.00$

c) $25.00 - 24.65$

d) $1.245 / 106.0$

4 Addition and subtraction

Rule for addition and subtraction

When adding or subtracting measured values, round the answer to the least precise decimal place in the calculation.

Worked example: Adding masses

A student combines three masses:

$$12.45 \text{ g} + 3.2 \text{ g} + 0.567 \text{ g} = 16.217 \text{ g}.$$

The least precise value is 3.2 g, which is given to one decimal place. Therefore:

$$\boxed{16.2 \text{ g}}.$$

The answer 16.217 g is over-precise because the value 3.2 g was not measured to the thousandth of a gram.

Worked example: Burette titre

A burette initial reading is 0.15 cm^3 , and the final reading is 24.80 cm^3 .

$$\text{titre} = 24.80 - 0.15 = 24.65 \text{ cm}^3.$$

Both readings have two decimal places, so the titre is reported to two decimal places:

$$\boxed{24.65 \text{ cm}^3}.$$

Quick exercise: Addition and subtraction practice

Round each answer correctly.

a) $18.45 \text{ g} + 0.6 \text{ g} + 2.317 \text{ g}$

b) $50.00 \text{ cm}^3 - 24.65 \text{ cm}^3$

c) $101.325 \text{ g} + 2.4 \text{ g} + 0.56 \text{ g}$

5 Multiplication and division

Rule for multiplication and division

When multiplying or dividing measured values, round the answer to the same number of significant figures as the least precise measured value.

Worked example: Simple multiplication

$$2.50 \times 3.1 = 7.75.$$

2.50 has 3 significant figures. 3.1 has 2 significant figures. Therefore the final answer should have 2 significant figures:

$$\boxed{7.8}.$$

Worked example: Titration concentration

A monoprotic acid is titrated with $0.1000 \text{ mol dm}^{-3}$ NaOH. The mean titre is 24.65 cm^3 , and the acid aliquot is 25.00 cm^3 . For a 1:1 reaction:

$$c_{\text{acid}} = \frac{c_{\text{NaOH}} V_{\text{NaOH}}}{V_{\text{acid}}} = \frac{0.1000 \times 24.65}{25.00} = 0.09860 \text{ mol dm}^{-3}.$$

Each measured value has 4 significant figures, so the final answer is:

$$\boxed{0.09860 \text{ mol dm}^{-3}}.$$

The final zero is significant. Writing $0.0986 \text{ mol dm}^{-3}$ would show only 3 significant figures.

Quick exercise: Multiplication and division practice

Round each answer correctly.

a) $\frac{2.50 \times 0.1050}{25.00}$

b) $\frac{1.245}{106.0}$

c) $\frac{0.1000 \times 23.70}{25.0}$

6 Do not round too early

Students often lose accuracy by rounding intermediate values too aggressively. Keep extra digits during calculation, then round the final result.

Bad habit

Do not copy a rounded intermediate value into the next line unless the teacher specifically asks for stepwise rounding. Use the calculator memory or keep several guard digits.

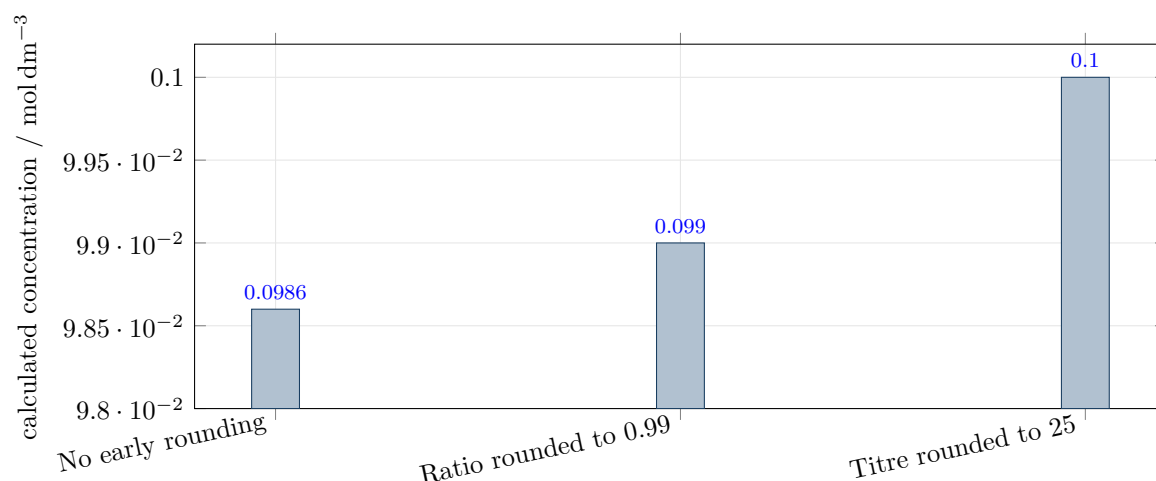


Figure 1: The same titration calculation can drift if intermediate values are rounded too early. The chemically defensible result comes from unrounded intermediate calculation followed by final rounding.

Worked example: Early rounding changes the answer

Correct calculation:

$$\frac{0.1000 \times 24.65}{25.00} = 0.09860.$$

If a student first rounds 24.65/25.00 to 0.99, then:

$$0.1000 \times 0.99 = 0.0990.$$

This is close, but it is less faithful to the original measurements. If the student rounds 24.65 to 25, the result becomes 0.1000, which is a much larger distortion.

Quick exercise: Spot the damage

A student calculates:

$$\frac{0.1000 \times 24.35}{25.00}.$$

a) Calculate the result without early rounding.

b) Now round 24.35/25.00 to two decimal places first, then multiply by 0.1000. What result do you get?

c) Which result should be reported? Explain.

7 Exact numbers do not limit significant figures

Some numbers are exact because they are counted, defined, or part of a chemical equation. Exact numbers do not reduce the number of significant figures in a final answer.

Exact numbers

The following are usually exact:

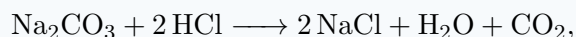
- stoichiometric coefficients in a balanced equation;

- counted objects, such as 3 trials or 5 tablets;
- defined unit conversions, such as $1000\text{ cm}^3 = 1\text{ dm}^3$.

Exact numbers do not control final rounding. Measured values do.

Worked example: Stoichiometric coefficient

For



the coefficient 2 is exact. It does not mean the final answer must have one significant figure. The limiting precision comes from measured masses, volumes, and concentrations.

Quick exercise: Exact or measured?

Classify each value as exact or measured.

Value	Exact or measured?
2 in $2\text{H}_2 + \text{O}_2 \longrightarrow 2\text{H}_2\text{O}$	
24.65 cm^3 titre from a burette	
$1000\text{ cm}^3 = 1\text{ dm}^3$	
5 tablets counted into a beaker	
1.245 g measured on a balance	

8 Using uncertainty to choose significant figures

The most mature way to choose significant figures is to use uncertainty. If the uncertainty is large, extra digits in the value are meaningless.

Value and uncertainty must match

A common rule is:

- 1) Round the uncertainty to 1 significant figure. Use 2 significant figures if the first digit is 1 or 2 and extra clarity is useful.
- 2) Round the measured value to the same decimal place as the rounded uncertainty.

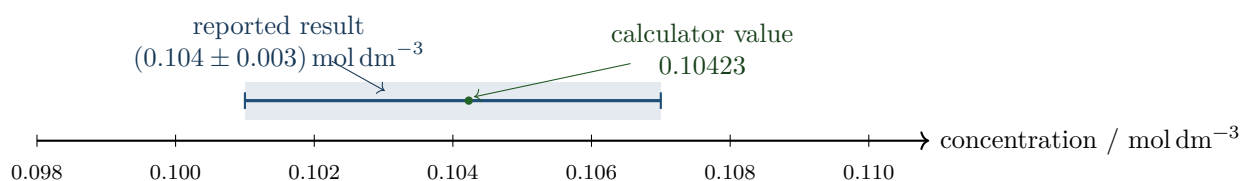


Figure 2: The uncertainty interval controls the useful decimal place. Reporting 0.10423 ± 0.003 would pretend that the final digits are meaningful, even though the uncertainty is much larger.

Worked example: Round using uncertainty

A calculation gives:

$$c = 0.104\,23\text{ mol dm}^{-3}, \quad u = 0.0031\text{ mol dm}^{-3}.$$

Round the uncertainty:

$$0.0031\text{ mol dm}^{-3} \approx 0.003\text{ mol dm}^{-3}.$$

Round the value to the same decimal place:

$$c = (0.104 \pm 0.003) \text{ mol dm}^{-3}.$$

Quick exercise: Round the result with uncertainty

Write each result in a sensible final form.

a) $12.376 \text{ g} \pm 0.428 \text{ g}$

b) $0.046981 \text{ mol dm}^{-3} \pm 0.00124 \text{ mol dm}^{-3}$

c) $104.728 \text{ mg dm}^{-3} \pm 2.6 \text{ mg dm}^{-3}$

9 pH and logarithms

pH is logarithmic, so it has a special significant-figure convention. The digits before the decimal point in pH indicate the power of ten. The decimal places in pH carry the precision information.

pH rule

For

$$\text{pH} = -\log_{10}[\text{H}^+],$$

the number of decimal places in pH should match the number of significant figures in $[\text{H}^+]$.

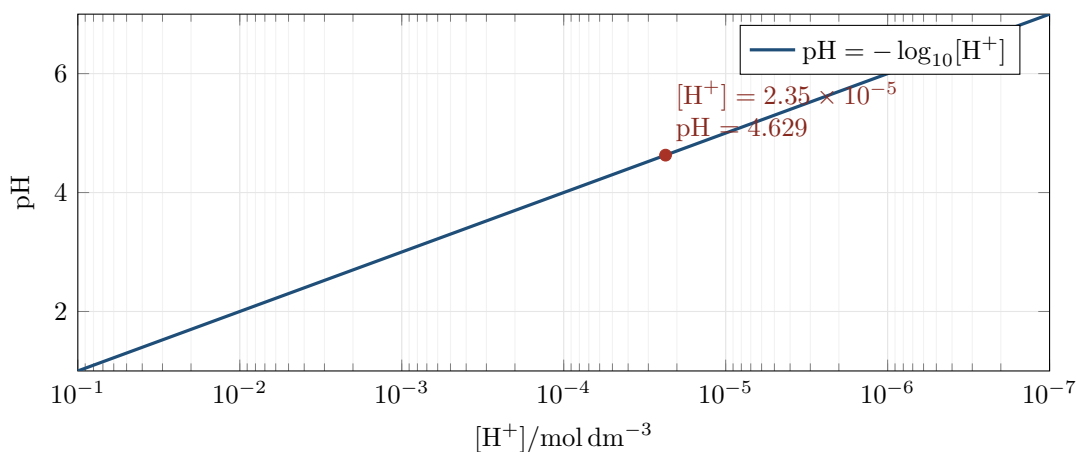


Figure 3: pH is logarithmic. Significant figures in $[\text{H}^+]$ become decimal places in pH.

Worked example: pH precision

$$[\text{H}^+] = 2.35 \times 10^{-5} \text{ mol dm}^{-3}.$$

The concentration has 3 significant figures. Therefore the pH should have 3 decimal places:

$$\text{pH} = 4.629.$$

Quick exercise: pH rounding

Calculate and round the pH correctly.

a) $[\text{H}^+] = 1.0 \times 10^{-3} \text{ mol dm}^{-3}$

b) $[\text{H}^+] = 3.16 \times 10^{-4} \text{ mol dm}^{-3}$

c) $[\text{H}^+] = 5 \times 10^{-6} \text{ mol dm}^{-3}$

10 Choosing significant figures in written conclusions

A final chemistry conclusion should be rounded and worded to match the evidence. Significant figures are part of scientific honesty.

Weak reporting	Stronger reporting
$c = 0.103847293 \text{ mol dm}^{-3}$	The concentration was reported as $(0.104 \pm 0.002) \text{ mol dm}^{-3}$.
pH = 3	The pH was 3.00, based on $[\text{H}^+] = 1.0 \times 10^{-3} \text{ mol dm}^{-3}$.
The titre was about 25.	The mean concordant titre was 24.65 cm^3 .
The result was accurate.	The result agreed with the accepted value within the stated uncertainty.

Reporting rule: Practical workflow

- 1) Record raw data with the precision of the instrument.
- 2) Keep extra digits during calculations.
- 3) Apply operation rules or uncertainty rules at the end.
- 4) Include units.
- 5) Do not report digits that the experiment cannot justify.

Quick exercise: Improve the reporting

Rewrite each result in a stronger form.

a) A calculator gives $0.098604392 \text{ mol dm}^{-3}$. The measured values all had 4 significant figures.

b) A result is 16.217 g , but one mass was measured only to 0.1 g .

c) A result is $0.104728 \text{ mol dm}^{-3} \pm 0.0026 \text{ mol dm}^{-3}$.

11 Mini-practical: reporting a titration result

A student titrates 25.00 cm^3 of acid with $0.1000 \text{ mol dm}^{-3} \text{ NaOH}$. The reaction ratio is 1:1.

Trial	Initial burette / cm^3	Final burette / cm^3	Titre / cm^3
1	0.20	24.90	
2	0.15	24.80	
3	0.10	24.75	
4	0.25	24.90	

Quick exercise: Complete the titration report

a) Calculate each titre and write it to the correct number of decimal places.

b) Calculate the mean titre.

c) Calculate the acid concentration using unrounded intermediate values.

d) Report the final concentration to a suitable number of significant figures.

e) Write one sentence explaining why your final number of digits is justified.

Answer key

Section 1

School burette 24.65 cm³: plausible. Measuring cylinder 23.47 cm³: not plausible if marked every 1 cm³. Balance 2.347 g: plausible for 0.001 g resolution. pH = 4.126: not plausible for a meter displaying only two decimals; report around pH = 4.13.

Section 2

Value	Significant figures	Value	Significant figures
0.00320	3	40.05	4
2500	ambiguous	2.500×10^3	4
0.1000	4	100.0	4
7.080	4	6.02×10^{23}	3

Section 3

(a) Addition rule: decimal places. (b) Multiplication/division rule: significant figures. (c) Subtraction rule: decimal places. (d) Division rule: significant figures.

Section 4

- a) $18.45 + 0.6 + 2.317 = 21.367 \rightarrow 21.4$ g.
- b) $50.00 - 24.65 = 25.35$ cm³.
- c) $101.325 + 2.4 + 0.56 = 104.285 \rightarrow 104.3$ g.

Section 5

- a) $(2.50 \times 0.1050)/25.00 = 0.01050 \rightarrow 0.0105$ because 2.50 has 3 significant figures.
- b) $1.245/106.0 = 0.011745... \rightarrow 0.01175$ with 4 significant figures.
- c) $(0.1000 \times 23.70)/25.0 = 0.0948$ with 3 significant figures because 25.0 has 3 significant figures.

Section 6

- a) $(0.1000 \times 24.35)/25.00 = 0.09740$ mol dm⁻³.
- b) $24.35/25.00 = 0.974$, rounded to two decimal places gives 0.97; $0.1000 \times 0.97 = 0.0970$ mol dm⁻³.
- c) Report the result from unrounded intermediate calculation, then round appropriately: 0.09740 mol dm⁻³ if all measured values have 4 significant figures.

Section 7

Exact: stoichiometric coefficient 2, 1000 cm³ = 1 dm³, 5 counted tablets. Measured: 24.65 cm³ titre, 1.245 g mass.

Section 8

- a) $12.376 \text{ g} \pm 0.428 \text{ g} \rightarrow 12.4 \text{ g} \pm 0.4 \text{ g}$.
- b) $0.046981 \text{ mol dm}^{-3} \pm 0.00124 \text{ mol dm}^{-3} \rightarrow (0.0470 \pm 0.0012) \text{ mol dm}^{-3}$ if using two significant figures for an uncertainty beginning with 1; otherwise $(0.047 \pm 0.001) \text{ mol dm}^{-3}$.
- c) $104.728 \text{ mg dm}^{-3} \pm 2.6 \text{ mg dm}^{-3} \rightarrow (105 \pm 3) \text{ mg dm}^{-3}$.

Section 9

- a) $[\text{H}^+] = 1.0 \times 10^{-3}$: 2 significant figures, so $\text{pH} = 3.00$.
b) $[\text{H}^+] = 3.16 \times 10^{-4}$: 3 significant figures, $\text{pH} = 3.500$.
c) $[\text{H}^+] = 5 \times 10^{-6}$: 1 significant figure, $\text{pH} = 5.3$.

Section 10

- a) $0.098604392 \rightarrow 0.098\,60\text{ mol dm}^{-3}$ for 4 significant figures.
b) $16.217\text{ g} \rightarrow 16.2\text{ g}$ if one value was measured to one decimal place.
c) $0.104\,728\text{ mol dm}^{-3} \pm 0.0026\text{ mol dm}^{-3} \rightarrow (0.105 \pm 0.003)\text{ mol dm}^{-3}$.

Mini-practical

Titres: 24.70, 24.65, 24.65, 24.65 cm^3 . Mean titre:

$$\bar{V} = \frac{24.70 + 24.65 + 24.65 + 24.65}{4} = 24.6625\text{ cm}^3.$$

For reporting a mean titre from burette readings, this can sensibly be rounded to 24.66 cm^3 or retained internally as 24.6625 for calculation. Concentration:

$$c_{\text{acid}} = \frac{0.1000 \times 24.6625}{25.00} = 0.098\,65\text{ mol dm}^{-3}.$$

A suitable final reported concentration is $0.098\,65\text{ mol dm}^{-3}$, using 4 significant figures from the measured values. If the mean titre is first rounded to 24.66 cm^3 , the concentration is $0.098\,64\text{ mol dm}^{-3}$; this illustrates why intermediate values should not be rounded unnecessarily.